

Analyzing the Effects of Geopolitical Risk on WTI Oil Prices for Strategic Decision Making

Background and Overview

West Texas Intermediate (WTI) is the primary benchmark for United States crude oil pricing and a key indicator for national fuel costs. WTI represents high quality, light and sweet crude oil produced domestically and priced at the Cushing, Oklahoma hub. Because it is the most actively traded crude benchmark in North America, WTI directly influences gasoline and diesel prices nationwide and plays a central role in refining economics and supply chain logistics.

Oil prices are driven by short-term changes in supply and demand, but they are also highly sensitive to geopolitical events. Conflicts, sanctions, political instability, and diplomatic shifts can disrupt supply routes, create uncertainty, and trigger rapid price movements. Recent tensions in the Middle East, including conflict involving Iran, have caused significant spikes in WTI prices and increased volatility across global energy markets. These developments directly affect organizations that rely on fuel, as rising oil prices increase operating costs and create financial uncertainty. This environment provides motivation for analyzing how different types of geopolitical events influence WTI price behavior.

Role and Motivation

In this project, we act as data analysts advising the organization on how geopolitical developments affect fuel costs. The broader problem we aim to solve is the difficulty organizations face in anticipating sudden price swings caused by global political events. These swings can dramatically increase operating costs, disrupt budgets, and reduce profitability.

By applying business analytics, we convert unpredictable geopolitical events into measurable patterns. This allows the organization to make informed decisions about hedging, procurement timing, and risk management. Addressing this issue is essential for maintaining financial stability in an environment where geopolitical volatility is becoming more frequent and more impactful.

Research Question

Our central question is: How do different categories of geopolitical events, including military escalation, geopolitical tension, policy uncertainty, and diplomatic de-escalation, affect WTI crude oil prices over short- and medium-term time horizons?

To answer this, we categorized geopolitical events into four regimes (geopolitical tension, policy uncertainty, military escalation, military deescalation), measured WTI price changes following each event, compared average price reactions across one-day, three-day, seven day, and thirty-day windows, and identified patterns that can guide organizational decision making. This question is specific, data driven, and directly tied to accessible datasets, making it appropriate for the SOAR analytics model.

Obtain the Data

The data used for the analysis was obtained through the Saint Louis Federal Reserve Economic Data website. It provided the WTI closing price of a barrel of oil for all days from January 1, 2021 until April 24, 2026, containing a total of 389 trading days. However, for the purpose of this analysis, it was decided to only consider the prices of oil since October 1, 2024. Additionally, it is important to note that the data omits weekend dates, given that major global futures exchanges, such as the NYMEX (where WTI future contracts are exchanged) are closed from Friday evening to Sunday evening EST. Because of this, in the data set, whenever percent changes are mentioned (daily, 3-day, 7-day etc.), they can be read best as x observations ago rather than actual calendar days.

Data Analysis

We began the analysis by researching key events that might impact the price of a barrel of oil and then flagging that event in our data set for future reference. We then discussed how we would like to classify and categorize each date whose oil price is based on the dominant backdrop influencing the oil market on that day, and we concluded that each trading day can be divided into one of the following four regimes: Geopolitical Tension, Policy Uncertainty, Military Escalation and De-escalation. Each regime starts at one of the flagged events and then runs until the next event triggers an appropriate change.

To answer the question of how different categories of geopolitical events impact WTI crude oil prices, we employed descriptive and diagnostic analytics as our primary methods of analyses. For each trading day, we calculated four price change metrics: the daily percent change, 3-day percent change- 7-day percent

change, and 30-day percent change, each measuring the price movement relative to the closing price a corresponding number of trading days prior. These metrics were chosen because oil markets do not react instantaneously to geopolitical events, the effects are typically reflected and materialized over days or weeks following a trigger event. Using a pivot table, we then computed the average daily, 3-day, 7-day, and 30-day percentage, price changes grouped by event regime to demonstrate possible patterns that are revealed across the chosen categories. Moreover, price frequency distribution tables were generated for the overall data set and for each individual category to observe any possible trend, and we calculated the mean, median, minimum, maximum, and standard deviation for each. Furthermore, data visualizations were created to demonstrate the effects of geopolitical events.

The primary limitation of the analysis is that regime classification is based on qualitative judgement rather than objective measurements, meaning the boundaries between categories involve some subjectivity. These regimes could also overlap in some dates, which was excluded in this analysis. Additionally, because oil prices are complex and influenced by many factors – production decisions, macroeconomic conditions, and currency fluctuations – the price changes observed with each category cannot be attributed exclusively to geopolitical events. Moreover, the sample size of our data set is relatively small, and some of the results might be more sensible to statistical outliers than others.

Consider the Future

The analysis shows that WTI crude oil responds differently depending on the type of geopolitical event, and these results can guide organizations in managing fuel related decisions.

Military escalation produces the strongest price increases, with WTI rising an average of 3.53 percent within three trading days, 8.57 percent within seven days, and 33.38 percent within thirty days. The price frequency distribution for this category also demonstrated high volatility, with its standard deviation of \$17.38 being nearly double that of the overall dataset's standard deviation of \$10.54. Because these events consistently lead to sharp spikes, organizations should increase hedging and secure fuel contracts early.

Diplomatic de-escalation has the opposite effect, with prices falling an average of 2.37 percent within seven trading days. However, the distribution tells a more nuanced story. It peaks sharply at \$65 with 30 days, and drops off quickly above \$70, meaning the bulk of de-escalation days traded in the \$60-\$65 range. The slightly elevated mean is likely being pulled up by a few outlier days at \$90–

\$100 that occurred during transition periods. The standard deviation of \$8.94 is the lowest of all four regimes, confirming that de-escalation brings the most price stability. During these periods, organizations benefit from delaying major fuel purchases until prices settle at lower levels.

Geopolitical tension without escalation leads to moderate increases, including a 3.73 percent rise over thirty days. Additionally, the mean for this category is close to the overall average, with the standard deviation of \$4.86 showing low volatility and change with stable prices. Staggered procurement is appropriate in these situations, allowing organizations to stay flexible while managing gradual upward pressure. Similar results are observed within the Policy Uncertainty category. The mean is \$4.79 below the overall average and a standard deviation of \$4.83. Policy uncertainty produces mixed results, so balanced hedging and close monitoring are recommended.

Future work could extend the analysis beyond thirty days to determine how long these effects last. Studying overlapping geopolitical events would also help clarify how markets react when tensions rise repeatedly. Industry specific analysis could provide more tailored guidance based on different levels of fuel exposure.

With more detailed data, such as real time geopolitical risk indices, shipping disruptions, refinery outages, global supply and demand balances, and intraday price movements, the analysis could become more precise. These datasets would help identify how quickly markets react to breaking news and which events create the strongest long-term effects. Combining richer data with the framework developed here would support more accurate forecasting and more effective fuel cost management.



